

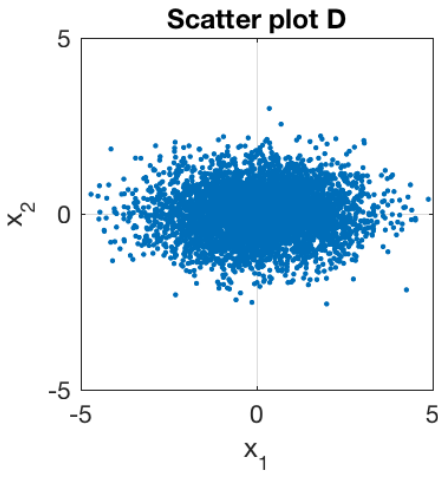
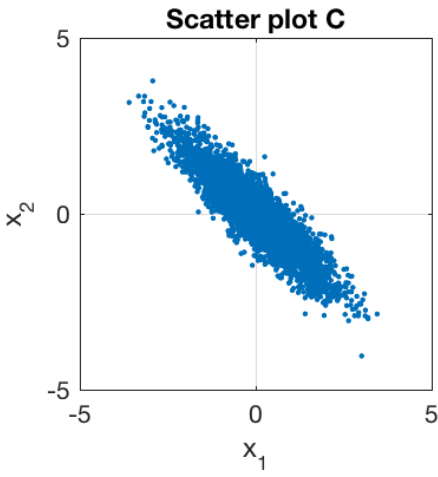
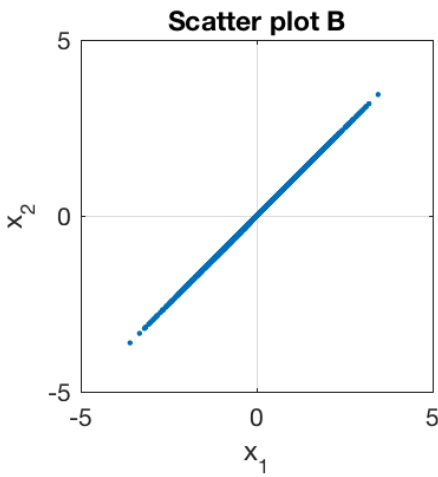
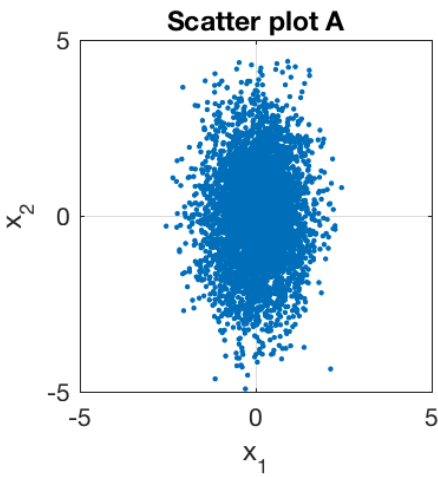
Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1. Consider the four scatter plots shown below. Each plot displays 5000 samples of a Gaussian random variable having zero mean and an unknown covariance.

1 / 1 point



Which one of these plots could have been created from sampling a Gaussian random vector having covariance matrix  $\Sigma = \begin{bmatrix} 2 & 0 \\ 0 & 0.5 \end{bmatrix}$ ?

- ☐ Scatter plot A
- ☐ Scatter plot B
- ☐ Scatter plot C
- ☒ Scatter plot D

✔ Correct  
Yes. This is the correct answer. Notice that  $x_1$  and  $x_2$  are uncorrelated and that  $x_1$  has greater variance than  $x_2$ .

2. Consider again the four scatter plots from Question 1. Which one of the plots could have been created from sampling a Gaussian random vector having covariance matrix  $\Sigma = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ ?

1 / 1 point

- ☐ Scatter plot A
- ☒ Scatter plot B
- ☐ Scatter plot C
- ☐ Scatter plot D

✔ Correct  
Yes. This is the correct answer. Notice that the correlation coefficient is 1, meaning that  $x_1$  is a deterministic function of  $x_2$ .

3. Consider again the four scatter plots from Question 1. Which one of the plots could have been created from sampling a Gaussian random vector having covariance matrix  $\Sigma = \begin{bmatrix} 1 & -0.9 \\ -0.9 & 1 \end{bmatrix}$ ?

1 / 1 point

- ☐ Scatter plot A
- ☐ Scatter plot B
- ☒ Scatter plot C
- ☐ Scatter plot D

✔ Correct  
Yes. This is the correct answer. Notice that the correlation coefficient is negative, so  $x_1$  and  $x_2$  are negatively related.